

SQL Server Inventory Management

From Database Design to Real-Time Stock Automation

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Why I Built This Project



Normalize Data

Eliminate redundant data and build a solid foundation with a fully normalized 5-table relational schema.



Automate Workflows

Enforce strict business rules and automate inventory updates directly at the database level.



Enterprise SQL

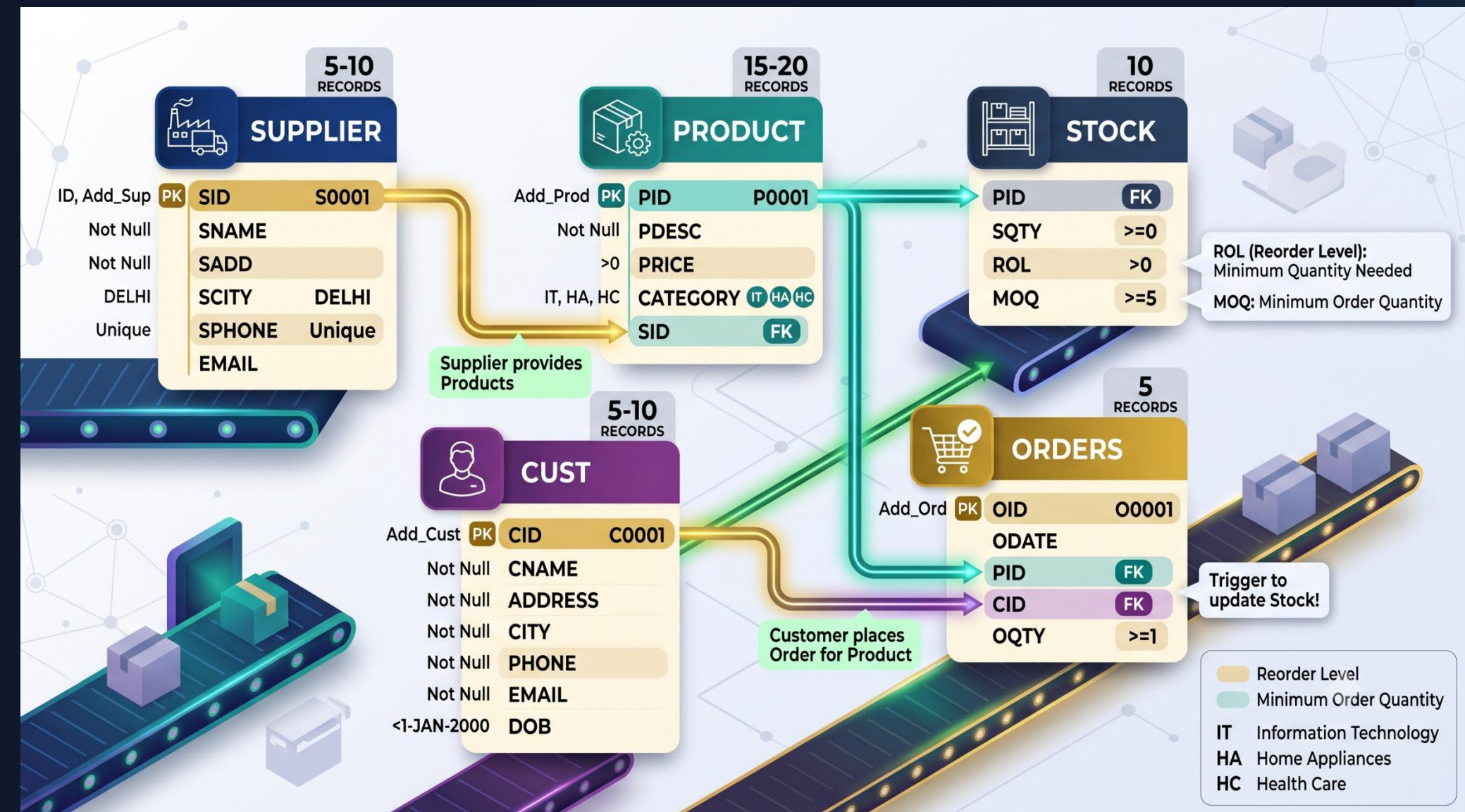
Move beyond basic queries to learn and implement production-style backend concepts and logical flows.

Relational Architecture

Core Entities & Relationships

The system connects 5 core entities using strict constraints to guarantee data integrity across all operations.

- Suppliers, Products, Stock, Customers, Orders
- Primary & Foreign Keys prevent orphaned records
- Customized CHECK, UNIQUE & NOT NULL constraints
- Clear one-to-many relationships for scalability



Backend Automation

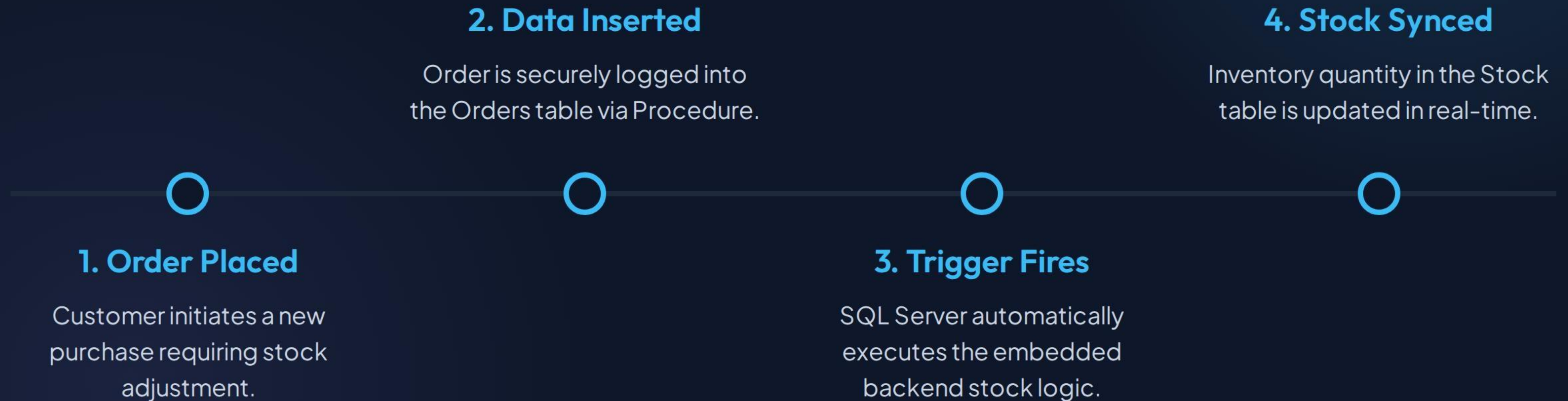
Procedures & Functions

Developed optimized Stored Procedures for modular, secure data entry. Created a custom SQL Function for automated, dynamic 5-character alphanumeric ID generation (e.g., P0021, C0011) utilizing SEQUENCES.

Real-Time Triggers

Engineered complex INSTEAD OF and AFTER triggers. These automatically calculate and synchronize stock quantities the exact moment new orders are placed, updated, or products are deleted.

Automated Stock Sync



Key SQL Competencies



Relational Database Design

Applied strict normalization rules (1NF to 3NF) and constructed a robust schema architecture.



Advanced Programmability

Built dynamic Stored Procedures and User Defined Functions for programmatic data handling.



Event-Driven Logic

Engineered complex DML Triggers (Insert/Update/Delete) for seamless background execution.



Data Governance

Enforced business rules natively within the database using advanced constraint methodology.

Version 2.0 Roadmap



Security Schema

Implement role-based access controls with specific execution permissions for administrators versus data entry clerks.



Dashboards

Connect the database to Power BI or Tableau to visualize real-time inventory levels, reorder alerts, and sales trends.



Audit Logs

Create historical tracking tables to automatically monitor and record all stock adjustments and order modifications over time.

Thank You!

Feedback and suggestions are always welcome.

Abhijeet Pandey | GIS | Python | SQL | Remote Sensing